CUES / QUESTIONS

1. Why does adverse yaw occur?
2. What are the 4 methods of reducing adverse yaw?
3. What is Dutch roll and how is it controlled?
4. What are the differences between a conventional tail and a T-tail?
5. What are the types of canards and how do they work?

NOTES

7. ADVERSE YAW





8. 4 METHODS OF REDUCING ADVERSE YAW

Rudder–Aileron Interconnect	Frise Ailerons	Differential Ailerons	Roll Spoilers
			
Rudder moves with ailerons.	Upward hinged portion creates additional drag.	Down aileron deflects more than up aileron.	Spoilers on up wing reduce lift and drag.

9. DUTCH ROLL



A coupled yawing and rolling motion.

Controlled by a yaw damper.

10. CONVENTIONAL TAIL vs T-TAIL

Conventional Tail



Conventional Tail – Pros & Cons

- Simpler design
- Lighter
- Less susceptible to wake/engine blast
- Can be more affected by wing wake

T-Tail



T-Tail – Pros & Cons

- Horizontal stabilizer above wing wake/engine blast
- Better control at high speeds
- Heavier and more complex
- Possible deep-stall recovery concerns

11. CANARD CONFIGURATIONS

Control Canard



Lifting Canard



Comparison

	Control Canard	Lifting Canard
Function	Provides pitch control	Provides lift + control
Lift contribution	Minimal	Significant
Potential issues	Nose-up tendency	Stall / stability concerns

SUMMARY – SECTION 2

- Adverse yaw is caused by unequal drag and can be reduced with several methods.
- Dutch roll is a coupled motion and is controlled by a yaw damper.
- Tail design affects stability and performance, with the T-tail offering benefits and trade-offs.
- Canards can be used for control or lift, each with different characteristics and considerations.